

Instruction Set & CPU Performance

1 Clock Cycles & Clock Rate

- **Clock Period (T):** Time duration of a single clock cycle (in seconds).
- **Clock Rate (f):** Number of clock cycles per second (in Hz).

$$\text{Clock Rate } (f) = \frac{1}{\text{Clock Period } (T)}$$

2 CPU Time

CPU Time refers to the actual time the CPU spends processing instructions.

$$\begin{aligned}\text{CPU Time} &= \text{CPU Clock Cycles} \times \text{Clock Cycle Time} \\ \text{or CPU Time} &= \frac{\text{CPU Clock Cycles}}{\text{Clock Rate}}\end{aligned}$$

3 CPI (Cycles Per Instruction)

- **Instruction Count (IC):** Total instructions executed.
- **CPI:** Average number of clock cycles per instruction.

$$\text{CPI} = \frac{\text{CPU Clock Cycles}}{\text{Instruction Count}} \quad \text{IPC (Instructions Per Clock)} = \frac{1}{\text{CPI}}$$

4 Final CPU Time Formula

$$\begin{aligned}\text{CPU Time} &= \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time} \\ \text{or CPU Time} &= \frac{\text{Instruction Count} \times \text{CPI}}{\text{Clock Rate}} = \frac{N \times \text{CPI}}{f}\end{aligned}$$

5 Multiple Instruction Classes

When using different instruction classes:

$$\text{Total CPU Clock Cycles} = \sum_{i=1}^n (\text{CPI}_i \times C_i)$$

Where:

- CPI_i = Cycles per instruction for class i
- C_i = Number of instructions of class i
- n = Number of instruction classes

6 Comparing Code Sequences

6.1 Instruction Class CPI

Class	CPI
A	1
B	2
C	3

6.2 Instruction Counts per Sequence

Sequence	A	B	C	Total Instructions
1	2	1	2	5
2	4	1	1	6

6.3 CPU Clock Cycles Calculation

Seq 1: $(2 \times 1) + (1 \times 2) + (2 \times 3) = 10$ cycles

Seq 2: $(4 \times 1) + (1 \times 2) + (1 \times 3) = 9$ cycles

6.4 CPI Comparison

$$\text{CPI}_1 = \frac{10}{5} = 2.0 \quad \text{CPI}_2 = \frac{9}{6} = 1.5$$

Conclusion: Sequence 2 is faster with a lower CPI.

7 Practice Problem: Clock Rate Calculation

Given:

- Computer A runs the program in 10s at 2 GHz
- Computer B should run it in 6s
- B uses 1.2x the clock cycles as A

7.1 Solution

Let C = clock cycles used by A.

$$\frac{C}{2 \times 10^9} = 10 \Rightarrow C = 20 \times 10^9$$
$$\text{For B: } \frac{1.2 \times C}{f_B} = 6 \Rightarrow f_B = \frac{1.2 \times 20 \times 10^9}{6} = \boxed{4 \text{ GHz}}$$

8 Factors Affecting CPU Performance

$$\text{CPU Time} = \frac{N \times \text{CPI}}{f}$$

- N = Instruction Count
- CPI = Cycles Per Instruction
- f = Clock Rate

8.1 Performance Influence Table

Component	Affects	Type
Algorithm	Instruction Count	Software
Programming Language	Instruction Count, CPI	Software
Compiler	Instruction Count, CPI	Software
Instruction Set Architecture	IC, CPI, Clock Rate	Both

9 Speedup Techniques

- **Cache Memory:** Reduces data access latency.
- **Pipelining:** Overlaps instruction stages for higher throughput.
- **Superscalar Architecture:** Executes multiple instructions in parallel.